

Inference-Aware Field Grading for Power Market Data Release: Amortized Subset Inferability and Budgeted Marginal Risk

Haocun Du

(affiliation to be completed)

Abstract

Power market operators publish large numbers of operational fields, and each field is typically screened for disclosure risk on its own. We show that this practice is unsafe: a field that looks harmless in isolation can become the decisive piece once an adversary holds one or two other published fields. We formalize release auditing as the estimation of *set-level inferability* $V_y(S)$, the out-of-sample R^2 that the strongest attacker in a fixed family can reach for a sensitive target Y_y using only the published field set S . To make the many subset queries of an audit affordable, we amortize them with a random-mask shared representation combined with a subset-specific closed-form ridge readout; against a shared output head this reduces the mean absolute error of V_y by $3.2\times$ and $4.6\times$ on two real markets. On top of V_y we define the *budgeted marginal inference capability* $M_{i\rightarrow y}^{(K)}$, the largest inference gain that field i can add over any background of at most K other candidate fields, and we characterize it as a budget-constrained marginal contribution index on a conditional evaluation function. We prove a truncated super-efficiency bound that yields a staged release budget, show that $M_{i\rightarrow y}^{(K)}$ is exactly the minimal threshold-independent safety margin, and prove that τ -criticality is equivalent to membership in a minimal unsafe set of size at most $K + 1$. On PJM and CAISO data with 41 candidate fields and 12 sensitive targets, exhaustive dedicated retraining shows that at $\tau = 0.7$ and $K = 2$, 35 of 41 (PJM) and 38 of 41 (CAISO) fields are safe alone but critical once at most two background fields are added. In a release audit over all size-three field sets, a single-field rule releases 19.1% (PJM) and 56.4% (CAISO) of the genuinely

unsafe sets, whereas the staged budget derived from our bound releases none while rejecting fewer than half of the safe sets. We further show that $M_{i \rightarrow y}^{(K)}$, being a maximum statistic, is inflated under weak attackers, and therefore report it together with a certified lower bound and a calibrated interval.

Keywords: data release auditing, inference risk, power market data, feature importance, amortized inference, marginal contribution

1. Introduction

Independent system operators publish extensive operational data: fuel-mix generation, load forecasts, ancillary-service requirements, locational marginal prices, and interchange schedules. Such transparency is mandated in most markets, yet the same data can be used to infer quantities the operator does not intend to disclose, for example total generation, metered load, or procured reserve volumes. Deciding which fields may be released is therefore a recurring operational problem.

Current practice screens fields one at a time. A candidate field is released if its correlation with a sensitive target is low, or if a model trained on that field alone predicts the target poorly. This paper starts from the observation that such screening is not merely imprecise but systematically unsafe in a specific way: it is blind to *combination risk*. On the two markets we study, at a threshold of $\tau = 0.7$ and a background budget of $K = 2$, **35 of 41 fields in PJM and 38 of 41 in CAISO are individually safe but become critical once an adversary additionally holds at most two other candidate fields**. A single-field release rule consequently releases 19.1% (PJM) and 56.4% (CAISO) of the field sets that a retrained attacker can actually exploit.

Making this quantitative requires answering two questions in sequence.

Q1: how inferable is an arbitrary field set? For a sensitive target Y_y and a candidate published set S , we need the out-of-sample inference capability $V_y(S)$ that an adversary could reach using only X_S . The direct route—retraining a dedicated attacker for every S —is exact but costly, and an audit issues many such queries: ad-hoc checks on proposed releases, exhaustive low-order sweeps, and searches for dangerous background combinations. We therefore ask whether a single target-agnostic pre-learning stage can serve a large number of different subset queries.

Q2: how does set-level inferability become a field-level security quantity? $V_y(S)$ alone does not grade fields. What a data owner needs is: *given that the adversary already holds some background information, how much can publishing field i increase the inferability of Y_y ?* We answer this with the budgeted marginal inference capability $M_{i \rightarrow y}^{(K)}$, together with the background that attains it—a *witness* that makes the risk auditable rather than merely numerical.

Contributions..

1. We formalize power-data release auditing as estimation of set-level inferability, separating three objects that are routinely conflated: the population optimum V_y^* , the empirical capability V_y^{ret} of a fixed attacker family, and the amortized estimate V_y^{amo} (Section 3).
2. We show that the right level at which to amortize is the *representation*, not the value. A random-mask shared backbone whose features feed a subset-specific closed-form ridge readout reduces V_y error by $3.2\times$ (PJM) and $4.6\times$ (CAISO) relative to a shared output head, and cuts the rate of dangerously underestimated sets from 33.6% and 28.7% to 12.6% and 3.8% (Sections 4 and 7).
3. We define $M_{i \rightarrow y}^{(K)}$ with an explicit public baseline B and adversary budget K , identify it as a budget-constrained marginal contribution index on the conditional evaluation function v_B , and prove three properties that the unconstrained index does not provide: a truncated super-efficiency bound that yields a staged release budget (Theorem 1), an exact characterization of $M_{i \rightarrow y}^{(K)}$ as the minimal threshold-independent safety margin (Theorem 2), and the equivalence between τ -criticality and membership in a minimal unsafe set of size at most $K + 1$ (Theorem 3) (Section 5).
4. On PJM and CAISO we compute exact $M^{(0..2)}$ for all 41 candidate fields and 12 targets by exhaustive dedicated retraining under a three-attacker family, quantify how much single-field screening misses, and show that the staged budget is the only rule among six that achieves zero dangerous release at a practical rejection rate (Section 8).
5. We document two failure modes of the point estimate and the corresponding remedy. The amortized estimator underestimates $M_{i \rightarrow y}^{(K)}$ for some fields, and $M_{i \rightarrow y}^{(K)}$ is a maximum statistic whose value is *inflated* when the attacker is weak—by 5–12% at $K = 2$ under a tenth of the

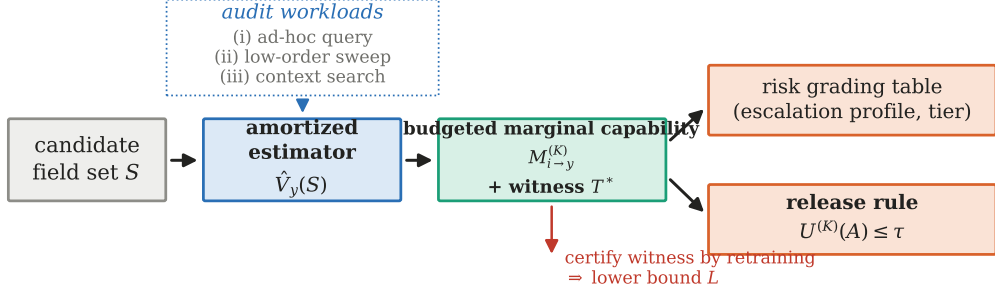


Figure 1: Audit pipeline. An arbitrary candidate field set S is scored by the amortized estimator $\hat{V}_y(S)$; sweeping low-order sets yields the budgeted marginal capability $M_{i \rightarrow y}^{(K)}$ together with a witness background; the field-level table then drives risk grading and a release rule. The three audit workloads that motivate amortization are marked.

auxiliary labels. Reporting a certified lower bound together with a calibrated upper bound removes both (Sections 7 and 8).

What we do not claim.. Random masking, closed-form ridge adaptation, maximum marginal contribution, and the notion of a witness context all predate this work; Section 2 attributes them. We also do not claim that amortization is necessary at the scale studied here: with 41 fields, exhaustive parallel retraining completes the full $K \leq 2$ audit in eleven minutes, and we report this explicitly in Section 7.

2. Related Work

2.1. Privacy and inference leakage in power data

Work on power-system privacy has concentrated on smart metering, where the concern is recovering household activity from fine-grained consumption traces [1, 2, 3]. Protection is usually achieved by perturbation or aggregation, with formal guarantees from differential privacy [4] or syntactic models such as k -anonymity [5]. These mechanisms answer a different question from ours. They bound what can be learned about an *individual record* after a release mechanism is applied, whereas a market operator releasing aggregate operational series needs to know which *fields* may be published at all, given that an adversary can train a supervised inference model on historical auxiliary labels. Our setting is closer to attribute inference than to membership

inference [6] or model inversion [7], since the adversary attacks the published data product rather than a deployed model.

2.2. Feature importance and marginal contribution

Shapley-based attribution [8, 9, 10, 11, 12] distributes a fixed total among features. This is the correct axiomatic answer for *fair allocation*, but it is the wrong object for disclosure risk: when two fields are near-duplicates, each receives roughly half the credit even though either one alone enables the inference. Catav et al. [13] address exactly this dilution with the Marginal Contribution Index (MCI), $I_\nu(f) = \max_S[\nu(S \cup \{f\}) - \nu(S)]$, prove it is the unique function satisfying marginal-contribution, elimination and minimalism axioms for monotone ν , and establish dummy, symmetry, super-efficiency, sub-additivity, a self-contribution lower bound, and duplication invariance. They also introduce the maximizing set as a *context*. Our $M_{i \rightarrow y}^{(K)}$ inherits all of this. What we add is the security-relevant parameterization—a public baseline B and an explicit adversary budget K —and the theory that this parameterization makes possible: the truncated bound and staged budget, the safety-margin characterization, and the link to minimal unsafe sets. We use Shapley and SAGE [10] only as controls, and interaction indices [14, 15] only to explain why $M^{(1)}$ exceeds $M^{(0)}$. Model-reliance measures [16, 17] are likewise controls: they describe what one fitted model uses, not what any attacker could achieve.

2.3. Amortized prediction and fast adaptation

Estimating $V_y(S)$ for many S is a set-conditioned prediction problem. Random feature masking during training is standard in models that must handle arbitrary conditioning patterns [18, 19, 20], and amortization has been used to avoid repeated per-instance optimization [21, 22], including amortized Shapley estimation [23, 24]. Closed-form ridge readouts on frozen features are the standard fast-adaptation mechanism in meta-learning [25, 26]. Our contribution is not any of these components but the finding, documented in Section 7, that *where* one amortizes decides whether the estimate is usable: sharing the representation while solving each subset’s readout in closed form is markedly more faithful than sharing the output head, and the difference is concentrated exactly on the high-marginal sets that matter for security.

3. Problem Formulation

3.1. Threat model

Two parties are involved.

The **auditor** is the data owner. It holds the complete historical record, including the sensitive targets, and performs the audit offline before release.

The **adversary** obtains the published fields and is assumed to possess historical auxiliary labels for the sensitive target, from which it trains an inference model. Risk is therefore *supervised inferability*, measured out of sample, not zero-shot inference. Section 8.5 studies how the conclusions change when the adversary’s auxiliary data is limited.

Let G be the set of candidate publishable fields, $B \subseteq G$ the *public baseline* already released (default $B = \emptyset$), and $H = G \setminus B$ the fields under audit. Let Y_y denote a sensitive target. The adversary’s *background budget* K is the number of fields from H , beyond B , that it is assumed to already hold.

3.2. Set-level inferability

We distinguish three objects that are often conflated.

Definition 1 (Population inferability). $V_y^*(S) = 1 - \inf_f \mathbb{E}[(Y_y - f(X_S))^2] / \text{Var}(Y_y)$, the infimum over all measurable f . It satisfies $V_y^*(\emptyset) = 0$, $0 \leq V_y^* \leq 1$, and is monotone in S , since additional fields may be ignored.

Definition 2 (Empirical inferability). Given data split into fit, val and test, and an attacker family $\mathcal{H}$, train each $h \in \mathcal{H}$ on X_S , select the attacker and its training epoch on val, and report test R^2 : $V_y^{\text{ret}}(S) = R_{\text{test}}^2(\hat{h}_S)$, $\hat{h}_S = \arg \max_{h \in \mathcal{H}, \text{epoch}} R_{\text{val}}^2(h; S)$.

Because of finite samples, optimization noise and selection noise, V_y^{ret} is not exactly monotone. An adversary holding S may, however, train on any subset $T \subseteq S$ and keep the best, so the operationally correct object is the *monotone closure*

$$\bar{V}_y(S) = V_y(\hat{T}_S), \quad \hat{T}_S = \arg \max_{T \subseteq S} R_{\text{val}}^2(T), \quad (1)$$

where the subset is chosen on validation data and the value is reported on test data. This repairs finite-sample non-monotonicity without letting the test set influence any selection. We write V_y for the closure throughout, and the closure applies equally to the amortized estimate V_y^{amo} defined in Section 4. Since $M_{i \rightarrow y}^{(K)}$ only involves sets of size at most $K + 1$, whose subsets are smaller still, the closure does not destroy the locality of $M_{i \rightarrow y}^{(K)}$.

3.3. Audit workloads

The reason V_y must be cheap is not the abstract size of $2^{|G|}$ but three concrete workloads: (i) ad-hoc queries, where a proposed release set S must be scored immediately; (ii) low-order exhaustive sweeps, where all sets up to a given size are scored to build a field-level risk table; and (iii) dangerous-context search, where one looks for backgrounds that sharply amplify a field. Section 7 reports where amortization does and does not pay off across these workloads.

3.4. Data-use boundary

For a given S , the closed-form readout uses only Y_y^{train} to obtain $\beta_{y,S}$; the test prediction fixes this parameter, and Y_y^{test} enters only the final R^2 . It never participates in backbone training, in solving β , in selecting the ridge parameter, in checkpoint selection, or in witness search. We state this explicitly because the auditor legitimately holds the sensitive labels, and the distinction between using them for fitting and using them for reporting is what makes the reported numbers out-of-sample.

4. Amortized Subset-Universal Inference Model

4.1. Overview

The model answers Q1: given a target Y_y and an arbitrary $S \subseteq G$, produce an estimate of $V_y(S)$ without retraining a dedicated attacker. It has two stages. A *random-mask pre-learning* stage trains one backbone that must predict the targets from randomly masked inputs, yielding a representation conditioned on which fields are visible. A *subset-specific adaptation* stage then solves, in closed form, a separate ridge readout for each queried (S, y) pair on top of frozen features.

Remark 1 (Scope of “universal”). *The model is universal over field subsets, which is what the audit workloads require. The backbone is shared across the 12 targets of a market; the readout is specific to both the subset and the target. Our experimental plan initially specified a separate backbone per target, but Section 7 shows that per-target backbones are uniformly worse and twelve times more expensive to pre-train, so the shared backbone is the main configuration and the per-target variant is reported as an ablation. Note that this concerns the estimator; the attacker family used to define ground truth does include single-target networks, because there the goal is to be as strong as possible, not to amortize.*

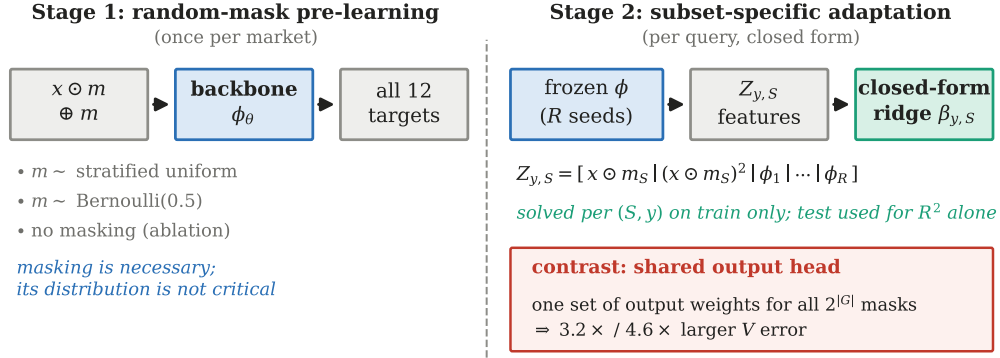


Figure 2: Two stages of the estimator. Pre-learning trains one backbone under random masks (left). At query time the backbone is frozen and only a subset-specific ridge readout is solved in closed form (right); the contrast is with a shared output head, which reuses one set of output weights for every conditioning pattern.

4.2. Random-mask pre-learning

Let $m \in \{0, 1\}^{|G|}$ be a visibility mask. The backbone $\phi_\theta(x \odot m, m)$ is a multilayer perceptron taking the masked input concatenated with the mask itself, trained to predict all sensitive targets under masks drawn from a sampling distribution. Concatenating m matters: without it, a zeroed field is indistinguishable from a field whose standardized value happens to be zero. We take ϕ to be the activation of the last hidden layer, a 256-dimensional vector.

The mask distribution is a design choice, and we report it as such. Section 7 compares a stratified uniform sampler, a Bernoulli(0.5) sampler and no masking at all; the first two are nearly indistinguishable while removing masking degrades the estimate. The honest summary is that random masking is necessary but its precise distribution is not critical, and we make no claim of a novel sampling scheme.

4.3. Subset-specific closed-form readout

For a query (S, y) with mask m_S , form the feature block

$$Z_{y,S} = [x \odot m_S \mid (x \odot m_S)^2 \mid \phi_{\theta_1}(x, m_S) \mid \dots \mid \phi_{\theta_R}(x, m_S)], \quad (2)$$

concatenating the masked raw inputs, their squares, and the frozen features of R independently seeded backbones ($R = 3$ in the main configuration).

The readout is ridge regression solved in closed form on the training split,

$$\beta_{y,S} = (Z_{\text{fit}}^\top Z_{\text{fit}} + \lambda I)^{-1} Z_{\text{fit}}^\top Y_y^{\text{fit}}, \quad (3)$$

with λ chosen on a fit/val partition *inside* the training split. The estimate is the test R^2 of $Z_{y,S}^{\text{test}} \beta_{y,S}$, after which the monotone closure (1) is applied. Solving (3) costs one small linear system, so a query is milliseconds rather than the tens of seconds a dedicated retraining requires.

4.4. Why not a shared output head

The alternative amortization—let one network map $(x \odot m, m)$ directly to the predicted target, and read $V_y(S)$ off its output—is strictly cheaper at query time and is what most masked-input models do. It is also substantially less faithful, and the gap is not uniform: it concentrates on sets with large marginal gains, which are precisely the ones that decide a release. Section 7 quantifies this. The mechanism is that a single set of output weights must serve all $2^{|G|}$ conditioning patterns, so the head interpolates across subsets; solving each subset’s readout separately removes this interference while keeping the expensive representation shared.

4.5. Error propagation to $M_{i \rightarrow y}^{(K)}$

Let $\tilde{V}$ be the closed estimate and $\tilde{\Delta}$ the induced marginals. Since the maximum is 1-Lipschitz in the supremum norm and a marginal is a difference of two values,

$$|\tilde{M}_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(K)}| \leq \max_{|T| \leq K} |\tilde{\Delta}_i(T) - \Delta_i(T)| \leq 2 \max_{|S| \leq K+1} |\tilde{V}(B \cup S) - V(B \cup S)|. \quad (4)$$

The practical consequence is that the quantity to control is the *marginal* error, not the value error: an estimator with a small bias in V can still misrank fields if its error is correlated across the two sets forming a marginal. We report marginal error explicitly.

5. Budgeted Marginal Inference Capability

5.1. Definition

Definition 3 (Budgeted marginal inference capability). *For target y , public baseline B , candidate set $H = G \setminus B$ and budget K ,*

$$M_{i \rightarrow y}^{(K)} = \max_{T \subseteq H \setminus \{i\}, |T| \leq K} [V_y(B \cup T \cup \{i\}) - V_y(B \cup T)]. \quad (5)$$

Any T^* attaining the maximum is a witness background.

Two derived quantities are used for interpretation only: the single-field capability $a_i^B = M_{i \rightarrow y}^{(0)} = V_y(B \cup \{i\}) - V_y(B)$, and the second-order interaction $I_{ij}^B = V_y(B \cup \{i, j\}) - V_y(B \cup \{i\}) - V_y(B \cup \{j\}) + V_y(B)$.

5.2. Relation to the marginal contribution index

Proposition 1. *Let $v_B(T) := V_y(B \cup T) - V_y(B)$ for $T \subseteq H$. Then (a) v_B is monotone, non-negative and satisfies $v_B(\emptyset) = 0$, hence meets all assumptions Catav et al. [13] impose on an evaluation function; (b) $M_{i \rightarrow y}^{(K)} = \max_{|T| \leq K} [v_B(T \cup \{i\}) - v_B(T)]$, i.e. $M_{i \rightarrow y}^{(K)}$ is the budget-constrained MCI of v_B ; (c) for $K \geq |H| - 1$, $M_{i \rightarrow y}^{(K)}$ equals the unconstrained MCI of v_B , and for $B = \emptyset$ the original MCI of V_y .*

Proof. (a) V_y is monotone and $B \cup T \subseteq B \cup T'$ whenever $T \subseteq T'$; also $v_B(\emptyset) = V_y(B) - V_y(B) = 0$. (b) Substituting the definition, $V_y(B \cup T \cup \{i\}) - V_y(B \cup T) = v_B(T \cup \{i\}) - v_B(T)$. (c) The budget constraint becomes vacuous, so the feasible set is all $T \subseteq H \setminus \{i\}$. $\square$

By Proposition 1, every property MCI enjoys for monotone ν —dummy, symmetry, sub-additivity, the self-contribution lower bound $M_{i \rightarrow y}^{(K)} \geq V_y(B \cup \{i\}) - V_y(B)$, duplication invariance, and the dilution argument against Shapley—transfers to $M_{i \rightarrow y}^{(K)}$ and is cited rather than claimed. What does not transfer is everything that depends on the budget being finite, and that is where our theory lies. We state three results; proofs of the remaining propositions are in Appendix A.

5.3. Truncated super-efficiency and the staged budget

Theorem 1 (Truncated super-efficiency). *For any $A \subseteq H$:*

- (a) *if $|A| \leq K + 1$, then $V_y(B \cup A) - V_y(B) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$;*
- (b) *define $U^{(K)}(\emptyset) = 0$ and $U^{(K)}(A) = \min_{j \in A} [U^{(K)}(A \setminus \{j\}) + M_{j \rightarrow y}^{(\min(|A|-1, K))}]$.
If $|A| \leq K + 1$, then $V_y(B \cup A) - V_y(B) \leq U^{(K)}(A) \leq \sum_{i \in A} M_{i \rightarrow y}^{(K)}$;*
- (c) *if $M_{i \rightarrow y}^{(K)} = M_{i \rightarrow y}^{(|A|-1)}$ for all $i \in A$, then (a) and (b) hold without the restriction $|A| \leq K + 1$.*

Proof. (a) Fix any order $i_1, \dots, i_m$ of A with $m = |A|$ and let $A_t = \{i_1, \dots, i_t\}$. Telescoping, $v_B(A) = \sum_{t=1}^m [v_B(A_t) - v_B(A_{t-1})]$. The t -th term is the marginal of i_t over a background of size $t - 1 \leq K$, hence at most $M_{i_t \rightarrow y}^{(K)}$. Only $v_B(\emptyset) = 0$ is used. (b) Induct on $|A|$: $v_B(A) = v_B(A \setminus \{j\}) + \Delta_j(B \cup A \setminus \{j\}) \leq U^{(K)}(A \setminus \{j\}) + M_{j \rightarrow y}^{(|A|-1)}$, then minimize over j ; the second inequality follows from monotonicity of $M^{(k)}$ in k (Proposition 3). (c) Replace each bound in (a) by $M^{(|A|-1)} = M^{(K)}$. $\square$

Remark 2. For $K \geq |H| - 1$ part (a) reduces to the super-efficiency of [13] on v_B . It is not a corollary of it: since $M_{i \rightarrow y}^{(K)} \leq \text{MCI}$, the truncated inequality is stronger, and needs its own proof, although the telescoping technique is the same. Parts (b) and (c) have no unconstrained counterpart. Part (b) is what we deploy as the staged budget release rule in Section 8: it is tighter than the additive budget of (a) because a set of size three need not charge every member at the full budget K .

5.4. $M_{i \rightarrow y}^{(K)}$ as the minimal threshold-independent safety margin

Theorem 2 (Safety margin). Fix i, y, B and K . Consider three notions of margin: a uniform marginal bound m satisfying $\Delta_i(B \cup T) \leq m$ for all $|T| \leq K$; a universal margin m satisfying “for all τ and all $|T| \leq K$: $V_y(B \cup T) \leq \tau - m \Rightarrow V_y(B \cup T \cup \{i\}) \leq \tau$ ”; and, for a fixed τ , $m_\tau = \inf\{m \geq 0 : \forall |T| \leq K, V_y(B \cup T) \leq \tau - m \Rightarrow V_y(B \cup T \cup \{i\}) \leq \tau\}$. Then (a) m is a uniform marginal bound $\iff m$ is a universal margin $\iff m \geq M_{i \rightarrow y}^{(K)}$, so $M_{i \rightarrow y}^{(K)}$ is the minimal universal margin; (b) $m_\tau \leq M_{i \rightarrow y}^{(K)}$ for every τ , and the inequality can be strict.

Proof. (a) The first equivalence is the definition of the maximum. Universal margin $\Rightarrow$ uniform bound: for any T take $\tau = V_y(B \cup T) + m$, giving $\Delta_i(B \cup T) \leq m$. Uniform bound $\Rightarrow$ universal margin: $V_y(B \cup T \cup \{i\}) = V_y(B \cup T) + \Delta_i \leq (\tau - m) + m = \tau$. (b) $M_{i \rightarrow y}^{(K)}$ satisfies the fixed- τ condition, so $m_\tau \leq M_{i \rightarrow y}^{(K)}$. Strictness: with $B = \emptyset$, $K = 1$, $V_y(\{i\}) = 0.1$, $V_y(\{j\}) = 0.5$, $V_y(\{i, j\}) = 0.9$, we get $M_i^{(1)} = 0.4$, while at $\tau = 0.95$ no admissible background can be pushed above τ by adding i , so $m_\tau = 0$. $\square$

Theorem 2 is why $M_{i \rightarrow y}^{(K)}$ belongs in a grading table: it is a property of the field that does not depend on the threshold eventually chosen. It also delimits the claim: for one fixed τ the necessary margin may be strictly smaller, so

using $M_{i \rightarrow y}^{(K)}$ as a release budget at a fixed threshold is conservative. We therefore do not describe $M_{i \rightarrow y}^{(K)}$ as “the minimal margin at a given τ ”.

5.5. Criticality and minimal unsafe sets

Definition 4. Under baseline B , a set $E \subseteq H$ is a τ -minimal unsafe set (MUS) if $V_y(B \cup E) > \tau$ while $V_y(B \cup E') \leq \tau$ for every $E' \subsetneq E$. A field i is K -background τ -critical if there exists T with $|T| \leq K$, $i \notin T$, such that $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$.

Theorem 3 (Criticality $\iff$ small MUS). If V_y is monotone and $V_y(B) \leq \tau$, then i is K -background τ -critical if and only if i belongs to some MUS of size at most $K + 1$.

Proof. ($\Leftarrow$) Let $E \ni i$ be a MUS with $|E| \leq K + 1$ and take $T = E \setminus \{i\}$. By minimality $V_y(B \cup T) \leq \tau$, and $V_y(B \cup E) > \tau$. ($\Rightarrow$) Suppose $V_y(B \cup T) \leq \tau < V_y(B \cup T \cup \{i\})$. Among the unsafe subsets of $T \cup \{i\}$ choose one, E , minimal under inclusion; it is a MUS with $|E| \leq K + 1$. If $i \notin E$ then $E \subseteq T$, and *monotonicity* gives $V_y(B \cup T) \geq V_y(B \cup E) > \tau$, a contradiction. $\square$

Monotonicity is not decorative here. With $V_y(\{1\}) = 0.8$, $V_y(\{3\}) = 0.2$, $V_y(\{1, 3\}) = 0.2$, $V_y(\{1, 2, 3\}) = 0.9$ and $\tau = 0.5$, field 2 crosses the threshold over background $\{1, 3\}$, yet every unsafe set containing field 2 also contains the unsafe proper subset $\{1\}$, so no MUS contains it. This is one of the reasons the monotone closure (1) is applied before any $M_{i \rightarrow y}^{(K)}$ is computed; on raw V_y^{ret} we observed a 0.2% violation rate, which the closure removes.

5.6. Closed form at $K = 1$ and the role of interaction

Proposition 2. $M_{i \rightarrow y}^{(0)} = a_i^B$ and $M_{i \rightarrow y}^{(1)} = a_i^B + \max(0, \max_{j \in H \setminus \{i\}} I_{ij}^B)$.

Proof. For $K = 0$ the only background is $\emptyset$. For $K = 1$ the background is $\emptyset$ or some $\{j\}$, and $\Delta_i(B \cup \{j\}) = V_y(B \cup \{i, j\}) - V_y(B \cup \{j\}) = a_i^B + I_{ij}^B$. $\square$

Hence a field is escalated by a background partner *exactly when* it is strictly super-additive with that partner. This also shows why a naive synergy score is not the right trigger: with $\text{syn}_2(ij) = V_y(\{i, j\}) - \max(a_i, a_j)$ one has $I_{ij} = \text{syn}_2 - \min(a_i, a_j)$, so $\text{syn}_2 > 0$ does not imply escalation.

Table 1: Datasets and threat-model settings.

	PJM	CAISO
Complete records	6,556	8,722
Train / val / test	4,589 (688 val) / 1,967	6,105 (916 val) / 2,617
Candidate fields $ H $	41	41
Sensitive targets	12	12
Public baseline B	$\emptyset$ (default); load forecasts (case study)	$\emptyset$
Background budget K	$\{0, 1, 2\}$ in the main text, $K = 3$ in Appendix B	
Threshold τ	$\{0.5, 0.7, 0.9\}$, main text 0.7	

5.7. From field risk to grading and release audit

The audit deliverable per target is a table containing, for each field, the escalation profile $(M^{(0)}, M^{(1)}, M^{(2)})$, the witness background attaining $M^{(2)}$, a risk tier, and the τ -critical flag. Because Theorem 1(b) bounds the inferability of a whole proposed release set by a staged sum of field-level budgets, the same table also supports a release rule: accept A if $U^{(K)}(A) \leq \tau$. Section 8 compares this rule against correlation screening, single-field screening, an additive budget, and a direct estimate of $V_y(A)$.

6. Experimental Setup

6.1. Data and protocol

We use one year of hourly operational data from two independent markets, PJM and CAISO (Table 1). Preprocessing removes constant columns and rows with missing values, then splits the records 70/30 into train and test with a fixed shuffle seed; 15% of train is held out as validation. Standardization uses train statistics only.

Fields that are exact or deterministic affine duplicates of another field are merged, since they would otherwise inflate the field count without adding information; near-duplicates ($r < 0.999$) are deliberately *kept*, because such redundancy is precisely what the method must handle. Visible copies of sensitive targets are removed. This leaves 41 candidate fields and 12 sensitive targets per market. The full field lists, the merge rules, data fingerprints and the frozen protocol are provided as supplementary material; the protocol was fixed before the reported experiments were run and was amended only three times, each amendment recorded with its empirical justification.

6.2. Ground truth and attacker family

Ground truth is exhaustive dedicated retraining. For every field set of size at most three (18,000 sets per market) and every target, we train attackers using only that set. The attacker family $\mathcal{H}$ contains a single-target network, a multi-target network, and a histogram gradient-boosting regressor [27]; the family maximum is taken after selecting attacker and epoch on validation, and the monotone closure (1) is then applied.

Including all three matters. At set size three, the tree is selected for 55.2% of (set, target) pairs in PJM and 43.0% in CAISO, and the single-target network for 35.3% and 50.5%. Using only the multi-target network—the natural choice if one reuses the estimator architecture—understates ground truth by 0.0267 (PJM) and 0.0155 (CAISO) R^2 at size three, which is enough to flip τ -critical decisions.

Dedicated retraining at this scale is feasible because the independent networks are trained simultaneously as a batched tensor operation: masked inputs are zeroed, so the first-layer weights of masked columns neither affect the output nor receive gradient, making the batched computation functionally equivalent to training each network on its own columns. This reduces the per-set cost from 21.8 s (PJM, sequential) to 56.6 ms at batch size 1024.

6.3. Metrics

For V_y we report mean absolute error, bias, Spearman correlation and the *dangerous-underestimation rate*, the fraction of (set, target) pairs whose estimate falls below the truth by more than 0.05. For $M_{i \rightarrow y}^{(K)}$ we report absolute error, Kendall correlation of the field ranking, risk-tier agreement, witness recall $P(L \geq M - 0.02)$ where L is the certified lower bound, and τ -critical recall/precision. For release rules we report the *dangerous release rate*, the fraction of genuinely unsafe sets that the rule accepts, and the false rejection rate.

7. Part A: Does Amortization Estimate V and M Reliably?

7.1. RQ-A1/A2: fidelity and the level of amortization

Table 2 compares estimators against the three-attacker ground truth on all sets of size at most three.

Three findings follow.

Table 2: Estimator comparison against the official three-attacker ground truth, all sets of size ≤ 3 . “Dangerous” is the fraction of (set, target) pairs underestimated by more than 0.05. Best in each column in bold.

Market	Estimator	V MAE	V bias	Dangerous	$M^{(2)}$ MAE	Witness recall
PJM	raw ridge	0.1230	−0.1220	34.9%	0.1185	0.155
	raw + x^2 ridge	0.0898	−0.0880	29.0%	0.0951	0.189
	random features	0.0441	−0.0380	17.9%	0.0832	0.220
	<i>shared output head</i>	0.0892	−0.0875	33.6%	0.1179	0.152
	ϕ ridge	0.0319	−0.0238	15.1%	0.0363	0.382
	per-target backbone	0.0403	−0.0349	19.2%	0.0526	0.362
	$\phi + x, x^2$ (3 seeds)	0.0276	−0.0180	12.6%	0.0297	0.352
CAISO	raw ridge	0.1041	−0.1040	40.4%	0.0664	0.281
	raw + x^2 ridge	0.0676	−0.0673	25.8%	0.0581	0.453
	random features	0.0204	−0.0175	5.3%	0.0340	0.474
	<i>shared output head</i>	0.0705	−0.0701	28.7%	0.0581	0.358
	ϕ ridge	0.0178	−0.0143	4.6%	0.0184	0.732
	per-target backbone	0.0230	−0.0209	6.8%	0.0230	0.587
	$\phi + x, x^2$ (3 seeds)	0.0153	−0.0106	3.8%	0.0141	0.742

The level of amortization decides usability (RQ-A2).. Holding the backbone fixed and replacing the shared output head by a frozen- ϕ plus subset-specific closed-form readout reduces V MAE from 0.0892 to 0.0276 in PJM ($3.2\times$) and from 0.0705 to 0.0153 in CAISO ($4.6\times$), while the dangerous underestimation rate falls from 33.6% to 12.6% and from 28.7% to 3.8%. The same reordering appears in $M^{(2)}$ error ($4.0\times$ both markets). This is the cleanest evidence in the paper and it is a statement about *where* to amortize, not about any individual component.

The learned representation is necessary but not sufficient.. Dropping ϕ and keeping only x, x^2 degrades V MAE by $2.8\times$ (PJM) and $3.8\times$ (CAISO). Random Fourier features recover part of the gap but remain clearly behind, and—importantly—their witness recall stays low, meaning they misidentify which background is dangerous even when their average error looks acceptable.

Per-target backbones are worse and much more expensive.. A separate backbone per target costs 342s (PJM) and 463s (CAISO) to pre-train versus 27–40s per seed for the shared backbone, and is uniformly worse on every

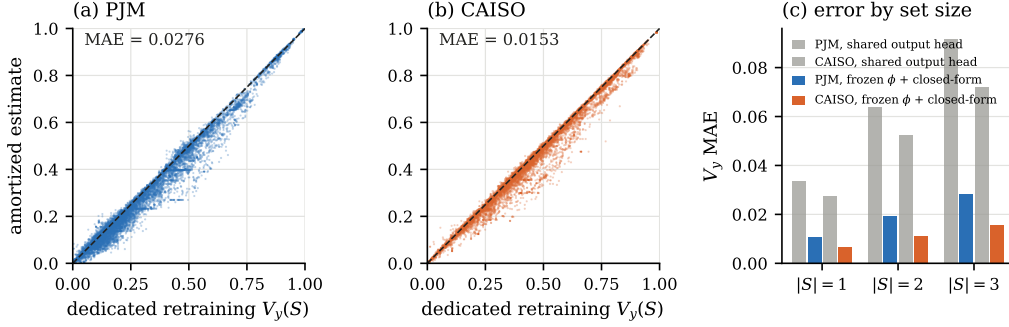


Figure 3: Fidelity of the amortized estimate against dedicated retraining, all sets of size ≤ 3 . (a)–(b) estimate versus truth; (c) error by set size for the shared output head versus the frozen-representation closed-form readout.

column. We therefore adopt the shared backbone, contrary to our own pre-registered plan, and record the deviation.

7.2. RQ-A3: is random masking necessary?

Comparing mask distributions used during pre-learning: no masking gives V MAE 0.0328 (PJM) / 0.0178 (CAISO), Bernoulli(0.5) gives 0.0308 / 0.0174, and the stratified uniform sampler gives 0.0318 / 0.0175. Masking helps, but the differences among distributions are within 0.002. The honest conclusion is that *random masking is necessary while its precise distribution is not critical*, and we make no claim to a superior sampler.

7.3. RQ-A4: reporting M as a certified interval

A point estimate of $M_{i \rightarrow y}^{(K)}$ is not the right output for a security audit. Precision of the estimated τ -critical flag is high (≥ 0.95 everywhere), but recall degrades with K , reaching 0.754 in PJM at $K = 2$, $\tau = 0.7$: a quarter of genuinely critical fields would be missed. We therefore report an interval.

The **lower bound** is certified. Any background $\tilde{T}$ proposed by the estimator yields $L_i = \Delta_i(B \cup \tilde{T}) \leq M_{i \rightarrow y}^{(K)}$ by definition, and evaluating Δ_i on ground truth makes L_i a genuine lower bound regardless of estimator quality. Certifying the estimator’s top three candidate backgrounds and keeping the best raises the certified ratio $\sum L / \sum M$ from 0.822 to 0.908 (PJM, $K = 2$) and from 0.953 to 0.974 (CAISO), and witness recall from 0.352 to 0.589 and from 0.742 to 0.886. Because the candidate backgrounds satisfy $|T| \leq K \leq 2$,

Table 3: Certified lower bound and calibrated upper bound for $M_{i \rightarrow y}^{(K)}$ (main estimator). Coverage is $P(L \leq M \leq U)$; the lower bound holds by construction.

Market	K	$\sum L / \sum M$ (top-1)	$\sum L / \sum M$ (top-3)	Cov. $\alpha=0.05$	Cov. $\alpha=0.01$	Width $U - L$
PJM	0	1.000	1.000	0.919	0.965	0.031
	1	0.898	0.960	0.927	0.982	0.074
	2	0.822	0.908	0.927	0.980	0.093
CAISO	0	1.000	1.000	0.931	0.961	0.019
	1	0.986	0.996	0.959	0.984	0.035
	2	0.953	0.974	0.935	0.986	0.040

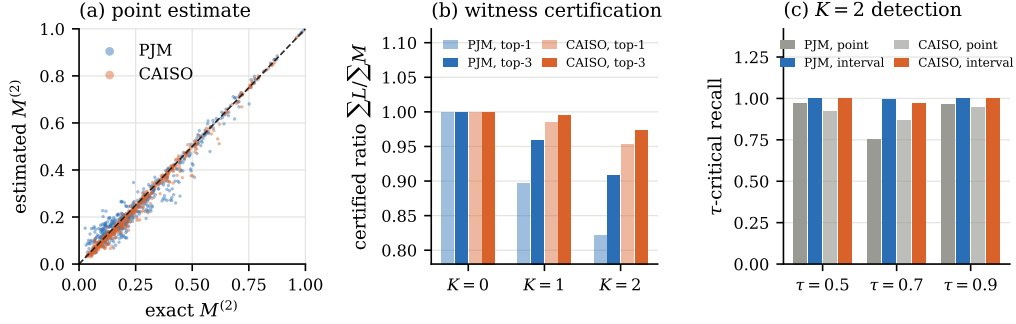


Figure 4: From point estimate to certified interval. (a) estimated versus exact $M^{(2)}$; (b) certifying the top-three candidate witnesses instead of only the best one raises the certified lower bound; (c) the interval-based conservative rule removes the recall deficit of the point estimate.

their values already lie in the exhaustive size- ≤ 3 truth table, so this costs no additional training.

The **upper bound** is calibrated. Following the conformal-style construction of [28], let q_α be the $1 - \alpha$ quantile of the underestimation $-(\tilde{\Delta} - \Delta)$ within a stratum of $\tilde{\Delta}$, estimated by *leave-one-field-out* cross-calibration so that the field being bounded never contributes to its own quantile; then $U_i = \max_T [\tilde{\Delta}_i(T) + q_\alpha(\text{stratum})]$. Table 3 reports coverage. At $\alpha = 0.01$ the interval $[L, U]$ covers the exact $M_{i \rightarrow y}^{(K)}$ for 98.0–98.6% of (field, target) pairs.

Used as a decision rule, the interval removes the recall problem. Flagging i as critical whenever some background satisfies $\hat{V}(T) - \delta_\alpha \leq \tau < \hat{V}(T \cup \{i\}) + \delta_\alpha$ raises recall at $K = 2$, $\tau = 0.7$ from 0.754 to 0.994 (PJM) and from 0.866 to 0.967 (CAISO), at the cost of precision falling to 0.870 and 0.826.

Table 4: Measured wall-clock cost on PJM, single GPU. “Full $K=2$ ” is 11,521 sets; “Full $K=3$ ” is 112,791 sets. Amortized columns include backbone pre-learning (27s per seed). The main estimator uses three seeds; the single-seed variant is the cheaper configuration reported in Table 2 as ϕ ridge.

Workload	Sequential retraining	Batched retraining	Amortized (3 seeds, main)	Amortized (1 seed)
Per query	21.8 s	56.6 ms	28.0 ms	2.4 ms
10^3 queries	6.1 h	1.6 min	1.8 min	0.5 min
10^5 queries	606 h	94 min	48 min	4.4 min
Full $K=2$	70 h	10.9 min	6.7 min	0.9 min
Full $K=3$	683 h	106 min	54 min	4.9 min
Break-even vs. batched	—	—	2,816 queries	495 queries

For a release audit this is the correct direction: false alarms are resolved by retraining the flagged candidate, whereas misses are not resolved at all. One caveat is recorded: at $\tau = 0.9$ the global quantile δ_α is too wide relative to the narrow band of V values near one, and precision falls to 0.420 (PJM) and 0.657 (CAISO); a stratified δ would be needed if $\tau = 0.9$ were the operating point.

7.4. RQ-A5: efficiency, and when amortization is not needed

Table 4 gives measured costs on a single GPU. The result that matters most is not the speedup but its scope: **at 41 fields the complete exact $K \leq 2$ audit takes about eleven minutes with batched dedicated retraining**, so amortization is not required to make this audit feasible. The break-even query count against batched retraining is 2,816 (PJM) and 3,082 (CAISO) for the three-seed main estimator, and 495 and 562 for the single-seed variant, which trades roughly 0.004 of V MAE for a $12\times$ cheaper query (Table 2).

Amortization becomes material in the regimes the audit workloads actually generate: a full $K = 3$ sweep (10^5 queries), larger field spaces (extrapolating by set count, a complete $K = 2$ audit at $p = 100$ comprises 166,750 sets, needing about 2.6 h of batched retraining versus 1.3 h for the three-seed estimator and 7 min for the single-seed variant), dangerous-context search over large backgrounds, and repeated recomputation when the baseline B , the data snapshot or the target changes. We state this scope explicitly rather than presenting amortization as the only feasible route.

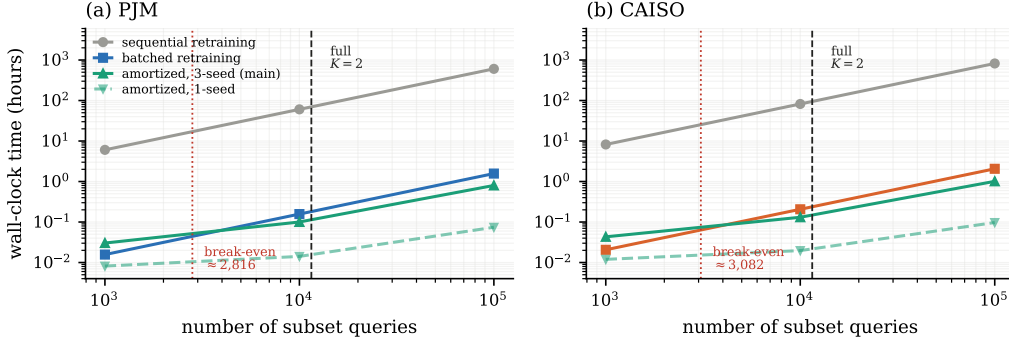


Figure 5: Cost of an audit as a function of the number of subset queries. The dashed vertical line marks the complete $K = 2$ sweep, which lies below the break-even point of the main estimator: at 41 fields, batched dedicated retraining is the cheaper way to obtain the exact table.

8. Part B: Is M the Right Field-Level Security Quantity?

8.1. RQ-B1: what single-field and attribution methods miss

Table 5 evaluates how well each score ranks fields by τ -criticality, using the exact $K = 2$ ground truth as the label.

Full-model attribution is clearly unsuited to this task: SAGE reaches only 0.594 on CAISO at $\tau = 0.7$, below even Pearson correlation. The reason is structural rather than statistical. These methods measure *model reliance*—what one fitted model happens to use—whereas disclosure risk is *potential inference capability* over all attackers. When several fields carry substitutable information, the fitted model commits to one and the attribution assigns the others near-zero importance, even though each alone supports the inference.

We also report an unfavorable result plainly. Exact single-field capability $M^{(0)}$ is a strong ranking baseline: at PJM $\tau = 0.7$ it reaches 0.877 against 0.888 for estimated $M^{(1)}$, a gap of 0.011. *On the ranking task, M is close to the best single-field baseline.* The advantage of M is not ranking; it is decision-making, which the next subsection isolates. Restricting the evaluation to fields that are critical only in combination ($K = 2$ critical but $K = 0$ safe) widens the gap slightly but does not change this conclusion.

8.2. RQ-B2/B3: escalation profiles and witnesses

Averaged over 41 fields and 12 targets, mean M rises from 0.144 ($K=0$) to 0.242 ($K=1$) to 0.287 ($K=2$) in PJM, and from 0.182 to 0.255 to 0.282

Table 5: PR-AUC for detecting τ -critical fields ($K = 2$). Exact $M^{(2)}$ is an oracle reference, not a deployable score.

Score	PJM			CAISO		
	$\tau=0.5$	$\tau=0.7$	$\tau=0.9$	$\tau=0.5$	$\tau=0.7$	$\tau=0.9$
Permutation importance (full model)	0.779	0.675	0.165	0.798	0.460	0.181
SAGE (full model)	0.816	0.730	0.283	0.832	0.594	0.307
Pearson $ r $	0.881	0.826	0.459	0.897	0.729	0.466
Spearman $ \rho $	0.895	0.829	0.459	0.895	0.747	0.422
Normalized mutual information	0.925	0.822	0.438	0.924	0.769	0.363
Single-field $M^{(0)}$ (exact)	0.907	0.877	0.480	0.916	0.764	0.454
Estimated $M^{(1)}$	0.939	0.888	0.601	0.941	0.823	0.532
Estimated $M^{(2)}$	0.939	0.877	0.573	0.944	0.829	0.549
<i>Exact $M^{(2)}$ (oracle)</i>	0.924	0.885	0.566	0.946	0.831	0.549

in CAISO. Most of the escalation therefore occurs at the *first* background field, consistent with Proposition 2: a single super-additive partner is usually enough.

The main text stops at $K = 2$. A $K = 3$ table exists but only under a weaker single-attacker family, and the discrepancy between that family and the official one is *larger than the increment it would illustrate*: at $K = 2$ the attacker-family gap is 0.0266 (PJM) and 0.0153 (CAISO), while the $K=2 \rightarrow K=3$ increment within the weaker family is 0.0213 and 0.0102. Reporting them side by side would be misleading, so $K = 3$ appears only in Appendix B. The same comparison independently supports stopping at $K = 2$: within a fixed family, $M^{(2)}/M^{(3)} = 0.924$ (PJM) and 0.963 (CAISO), so additional budget buys less than switching to a stronger attacker.

Witnesses make a risk statement auditable: each flagged field comes with a concrete background whose effect can be verified by retraining. We caution against over-reading them. Under an independent discovery/audit split, the *identity* of the witness reproduces only 12–37% of the time at $K \geq 1$, while the field’s risk level and τ -critical status reproduce at 0.78–1.00. A witness should therefore be presented as *one verifiable example* of a dangerous background, not as the unique dangerous combination.

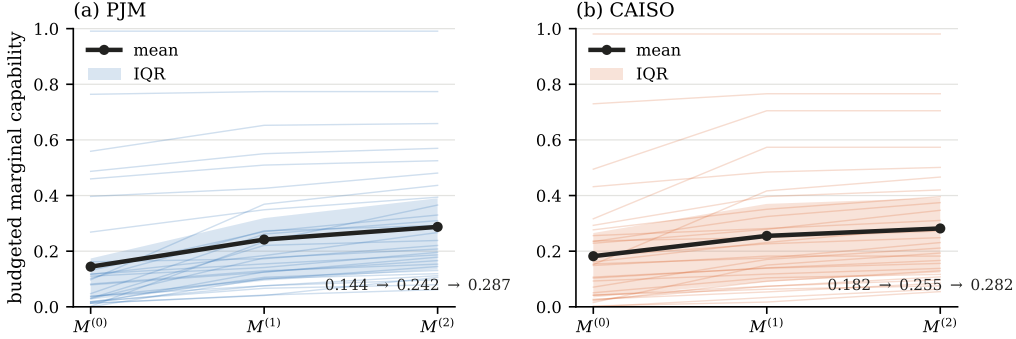


Figure 6: Inference escalation profiles $M^{(0)} \rightarrow M^{(1)} \rightarrow M^{(2)}$ under the official attacker family. Thin lines are individual (field, target) pairs; the shaded band is the interquartile range. Most escalation occurs at the first background field.

8.3. RQ-B4: risk is relative to what is already public

Taking B to be PJM’s two publicly released load forecasts changes the picture qualitatively. For `total_gen` and `metered_load`, $V_y(B)$ is already 0.974 and 0.991: the baseline alone leaks the target, and grading individual fields for these targets is meaningless. For other targets the maximum $M^{(2)}$ drops sharply—for `total_lmp_da` from 0.983 to 0.406—while reserve-type targets are almost unaffected. The Kendall correlation between field rankings with and without B ranges from 0.47 to 0.87. Disclosure risk is thus a property of a field *relative to the current public baseline*, not an intrinsic attribute, which is why B is explicit in Definition 3.

8.4. RQ-B5: release audit

This is the decision-level comparison. For a representative target per market (`total_gen` for PJM, `actual_load` for CAISO) we draw 2,000 random size-three candidate release sets and apply six rules at $\tau = 0.7$ (Table 6). A set is *unsafe* if the retrained attacker family exceeds τ on it.

The single-field rule—current practice—accepts 19.1% of unsafe PJM sets and 56.4% of unsafe CAISO sets while never rejecting a safe one. It is not conservative; it is blind in one direction. The additive budget of Theorem 1(a) eliminates dangerous releases but rejects 69–89% of safe sets, which no operator would deploy. The staged budget $U^{(2)}$ of Theorem 1(b) is the only rule achieving zero dangerous release at a rejection rate below one half. The direct amortized estimate has almost no false rejections but still admits 2.6–6.6% dangerous releases, making it a good *pre-filter* and a poor final gate.

Table 6: Release audit, 2,000 random size-three sets, $\tau = 0.7$. Dangerous release is the fraction of unsafe sets that the rule accepts.

Rule	PJM		CAISO	
	Dangerous rel.	False rej.	Dangerous rel.	False rej.
Single-field capability $\leq \tau$	19.1%	0.0%	56.4%	0.0%
Correlation ≤ 0.5	2.1%	37.9%	3.2%	22.6%
Direct estimate $\hat{V}(A) \leq \tau$	6.6%	0.0%	2.6%	0.1%
Staged budget $U^{(2)}(A) \leq \tau$	0.0%	49.6%	0.0%	32.8%
Additive budget $\sum \hat{M} \leq \tau$	0.0%	89.4%	0.0%	69.4%
Additive budget $\sum M \leq \tau$ (exact)	0.0%	86.3%	0.0%	74.0%

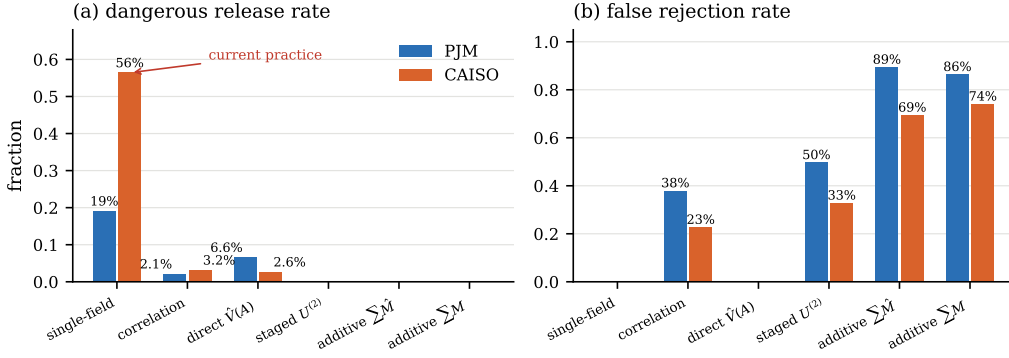


Figure 7: Release audit over 2,000 random size-three sets at $\tau = 0.7$. The single-field rule never rejects a safe set but accepts a large share of unsafe ones; the staged budget accepts none at a rejection rate below one half.

A practical deployment combines them: screen with $U^{(2)}$, and re-examine rejected sets with the direct estimate plus certified retraining.

The corresponding field grading is summarized in Table 7. The last column is the paper’s opening claim, now measured.

8.5. RQ-B6: how much auxiliary data does the adversary need?

All results so far assume the adversary trains on the same amount of data as the auditor. We relax this by giving the adversary a nested 10%, 25%, 50% or 100% of the auxiliary labels, with training *and* model selection confined to that subset.

Table 7: Field grading at $\tau = 0.7$, $K = 2$, representative target per market.

Market	Fields	$K=2$ critical	Critical alone	Safe alone, critical with ≤ 2
PJM	41	40	5	35
CAISO	41	41	3	38

Inferability degrades gracefully.. At 10% (459 PJM / 610 CAISO records) the mean V_y over size-three sets retains 80.7% and 90.2% of its full-data value. Data scarcity is not an effective defense.

Auditing with a weak attacker is unsafe.. If the auditor evaluates and releases using a 10% attacker while the real adversary has full data, 25.6% (PJM, $\tau=0.5$) and 13.2% (CAISO) of genuinely exploitable sets are released; τ -critical recall falls to 0.565 (PJM, $\tau=0.7$). Precision stays at 0.89–1.00, so the error is purely one-directional. This is the empirical argument for taking the maximum over an attacker family rather than fixing one attacker.

$M_{i \rightarrow y}^{(K)}$ is inflated, not deflated, under weak attackers.. Counterintuitively, although V_y is uniformly underestimated, $M^{(1)}$ and $M^{(2)}$ exceed their full-data values by 5–12% (Table 8), while $M^{(0)}$ remains monotone. The mechanism is that $M_{i \rightarrow y}^{(K)}$ is a *maximum* statistic: noisy marginals combined with a maximum over backgrounds systematically select positive noise. Decomposing $M_{\text{frac}} - M_{\text{full}} = \underbrace{[M_{\text{frac}} - \Delta_{\text{frac}}(T_{\text{full}})]}_{\text{selection gain}} - \underbrace{[M_{\text{full}} - \Delta_{\text{frac}}(T_{\text{full}})]}_{\text{genuine underestimation}}$ confirms this: the

selection gain grows with the number of candidate backgrounds—identically zero at $K=0$ (one candidate), 0.021 at $K=1$ (41 candidates) and 0.081 at $K=2$ (821 candidates) for PJM at 10%. Consistently, the witness selected at 10% agrees with the full-data witness only 11.6% of the time and retains only 77.4% of its effect.

This has a direct methodological consequence: **point estimates of $M_{i \rightarrow y}^{(K)}$ must not be compared across sample sizes or attacker strengths**, and any reported $M_{i \rightarrow y}^{(K)}$ should be accompanied by the certified lower bound of Section 7.3. Note that this is a *second, mechanically distinct* reason for the interval—Section 7.3 addresses estimator error, whereas the selection bias here would arise even with a perfect estimator applied to a weaker attacker.

Table 8: Attacker-capability sensitivity. M values are ratios to the full-data value; the selection-gain decomposition is at $K = 2$.

Market	Aux. labels	V_y ratio	$M^{(0)}$	$M^{(1)}$	$M^{(2)}$	Sel. gain ($K=2$)
PJM	10% (459)	0.807	0.855	0.923	1.077	0.081
	25% (1147)	0.907	0.928	0.998	1.124	0.060
	50% (2294)	0.958	0.974	0.984	1.023	0.025
	100% (4589)	1.000	1.000	1.000	1.000	0.000
CAISO	10% (610)	0.902	0.927	0.979	1.053	0.041
	25% (1526)	0.963	0.965	1.013	1.046	0.021
	50% (3052)	0.989	0.987	1.009	1.018	0.010
	100% (6105)	1.000	1.000	1.000	1.000	0.000

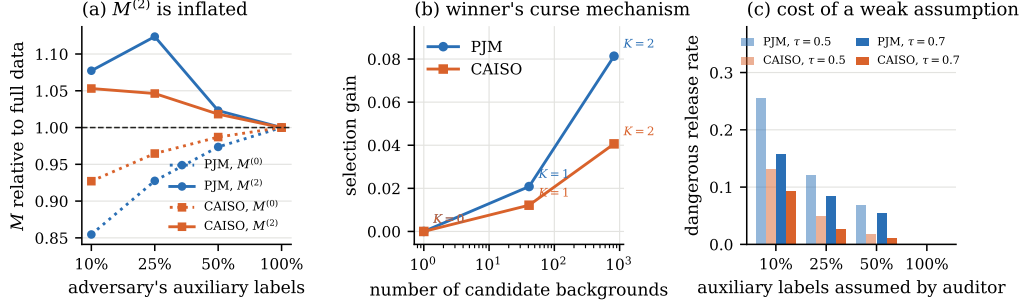


Figure 8: Attacker-capability sensitivity. (a) $M^{(2)}$ exceeds its full-data value when the adversary is weak, while $M^{(0)}$ does not; (b) the selection gain grows with the number of candidate backgrounds, identifying a maximum-statistic bias; (c) auditing under a weak attacker assumption releases sets that a full-data attacker can exploit.

9. Discussion

When is amortization worth it? Not at $p = 41$ with $K \leq 2$, where exact batched retraining takes eleven minutes. It becomes worthwhile at 10^5 -query workloads, larger field spaces, interactive auditing, and whenever the baseline, data snapshot or target changes and the whole table must be recomputed. We prefer to state this boundary rather than motivate the method with an infeasibility that does not hold at the scale we can verify.

Ranking versus deciding.. Exact single-field capability ranks fields nearly as well as M (Table 5). The value of M appears at the decision layer, where a rule must accept or reject a concrete set (Table 6). This distinction should be

kept explicit: a reader comparing only PR-AUC would reasonably conclude that the simpler score suffices.

Two biases, one remedy. The estimator underestimates M for some fields; a weak attacker inflates it for many. The mechanisms are unrelated but the remedy is the same—report $[L, U]$, with L certified by retraining the proposed witnesses. This is also why we retain dedicated retraining in the pipeline rather than replacing it: the amortized model generates candidates, and retraining certifies them.

Limitations. (i) We evaluate under random splits and do not assess temporal extrapolation; distribution drift may change the numerical values of M , although the audit framework itself is independent of how the data is split. (ii) Witness identity does not reproduce across independent data splits, so witnesses are examples, not unique certificates. (iii) The upper bound’s coverage relies on an exchangeability assumption across held-out fields and is not a finite-sample guarantee. (iv) For backgrounds of size four and above we have lower bounds only. (v) The threshold τ is exogenous; we report three values and their sensitivity rather than advocating one. (vi) The interval’s conservative decision rule loses precision at $\tau = 0.9$. (vii) The attacker family, though maximized over three members, excludes XGBoost and LightGBM variants and random forests; a stronger family would raise ground truth and, by the logic of Section 8.5, could only increase the measured combination risk. (viii) The sensitivity study of Section 8.5 uses a single attacker and one seed per fraction.

10. Conclusion

Screening power-market fields one at a time is not a conservative approximation of a combination-aware audit; it fails in the unsafe direction. Formalizing the audit as estimation of set-level inferability, and converting it into a field-level quantity through the budgeted marginal inference capability $M_{i \rightarrow y}^{(K)}$, makes the gap measurable: on two markets, 35 of 41 and 38 of 41 fields are safe alone and critical with at most two background fields, and a single-field release rule admits 19.1% and 56.4% of genuinely unsafe sets. The truncated super-efficiency bound turns the field-level table into a staged release rule that admits none of them while rejecting fewer than half of the safe sets. Two independent biases in the point estimate—estimator error and the selection bias of a maximum statistic under weak attackers—are

both handled by reporting a certified lower bound with a calibrated upper bound. Extending the evidence to field spaces of a hundred or more, and to a third market, is the natural next step.

Appendix A. Remaining proofs

Proposition 3 (Ladder, monotonicity in the candidate set, duplication invariance). (a) $M_{i \rightarrow y}^{(0)} \leq M_{i \rightarrow y}^{(1)} \leq \dots \leq M_{i \rightarrow y}^{(|H|-1)}$; (b) *enlarging the candidate set from H to $H' \supseteq H$ cannot decrease $M_{i \rightarrow y}^{(K)}$* ; (c) *if f' is a duplicate of f , i.e. $V_y(B \cup T \cup \{f'\}) = V_y(B \cup T \cup \{f\}) = V_y(B \cup T \cup \{f, f'\})$ for all T , then deleting f' leaves $M_{i \rightarrow y}^{(K)}$ unchanged for every $i \neq f, f'$.*

Proof. (a) and (b) are maxima over larger feasible sets. (c) Let T be a witness for i containing f' . If $f \notin T$, replace T by $(T \setminus \{f'\}) \cup \{f\}$: same size, same value. If $f \in T$, replace T by $T \setminus \{f'\}$: smaller size, same value. Either replacement stays within budget, so the maximum does not decrease after deleting f' ; by (b) it does not increase either. $\square$

Proposition 4 (Field-level bound on irreducible synergy). *Define the irreducible gain of S as $\text{syn}^B(S) = V_y(B \cup S) - \max_{T \subsetneq S} V_y(B \cup T)$. For $|S| \geq 2$, $\text{syn}^B(S) \leq \min_{j \in S} M_{j \rightarrow y}^{(|S|-1)}$.*

Proof. $\max_{T \subsetneq S} V_y(B \cup T) \geq V_y(B \cup S \setminus \{j\})$, hence $\text{syn}^B(S) \leq \Delta_j(B \cup S \setminus \{j\}) \leq M_{j \rightarrow y}^{(|S|-1)}$. $\square$

Remark 3. *Proposition 4 fails if strength is measured by the Harsanyi dividend [29]: with $V_y(\{i\}) = 0.5$ for $i = 1, 2, 3$, $V_y(\{i, j\}) = 0.5$ and $V_y(\{1, 2, 3\}) = 1.0$, the dividend of the triple is 1.0 while $M^{(2)} = 0.5$. This is why we define strength as the irreducible gain.*

Remark 4 (Escalation and submodularity). *Let $\Gamma_i^{(K)} := M_{i \rightarrow y}^{(K)} - M_{i \rightarrow y}^{(0)}$. Then $\Gamma_i^{(K)} = 0$ if and only if $\Delta_i(B \cup T) \leq \Delta_i(B)$ for all $|T| \leq K$. This is strictly weaker than submodularity: submodularity implies no escalation, but not conversely. A three-field monotone counterexample is $V_y(\{1\}) = 0.8$, $V_y(\{2\}) = 0.6$, $V_y(\{3\}) = 0.8$, all pairs 0.8, and $V_y(\{1, 2, 3\}) = 1.0$: every field has $M^{(2)} = a$, yet $\Delta_1(\{3\}) = 0 < \Delta_1(\{2, 3\}) = 0.2$. Hence “escalation $\iff$ synergy” is not a valid reading.*

Appendix B. $K = 3$ under the reduced attacker family

Exhaustive size-four retraining is available only for the multi-target network. Mean M under that family is 0.1346/0.2262/0.2608/0.2821 for PJM at $K = 0, 1, 2, 3$ and 0.1766/0.2455/0.2664/0.2766 for CAISO. The corresponding values under the official three-attacker family at $K \leq 2$ are 0.1443/0.2417/0.2873 (PJM) and 0.1820/0.2550/0.2817 (CAISO). Since the family gap at $K = 2$ (0.0266 / 0.0153) exceeds the $K=2 \rightarrow K=3$ increment within the weaker family (0.0213 / 0.0102), these columns must not be read as a continuation of the main-text escalation curve.

References

- [1] P. McDaniel, S. McLaughlin, Security and privacy challenges in the smart grid, *IEEE Security & Privacy* 7 (3) (2009) 75–77.
- [2] M. R. Asghar, G. Dán, D. Miorandi, I. Chlamtac, Smart meter data privacy: A survey, *IEEE Communications Surveys & Tutorials* 19 (4) (2017) 2820–2835.
- [3] G. Giaconi, D. Gündüz, H. V. Poor, Privacy-aware smart metering: Progress and challenges, *IEEE Signal Processing Magazine* 35 (6) (2018) 59–78.
- [4] C. Dwork, A. Roth, The algorithmic foundations of differential privacy, *Foundations and Trends in Theoretical Computer Science* 9 (3–4) (2014) 211–407.
- [5] L. Sweeney, k-anonymity: A model for protecting privacy, *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* 10 (5) (2002) 557–570.
- [6] R. Shokri, M. Stronati, C. Song, V. Shmatikov, Membership inference attacks against machine learning models, in: *IEEE Symposium on Security and Privacy (S&P)*, 2017, pp. 3–18.
- [7] M. Fredrikson, S. Jha, T. Ristenpart, Model inversion attacks that exploit confidence information and basic countermeasures, in: *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2015, pp. 1322–1333.

- [8] L. S. Shapley, A value for n-person games, *Contributions to the Theory of Games* 2 (28) (1953) 307–317.
- [9] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems* 30 (NeurIPS), Long Beach, USA, 2017, pp. 4765–4774.
- [10] I. Covert, S. M. Lundberg, S.-I. Lee, Understanding global feature contributions with additive importance measures, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33, 2020, pp. 17212–17223.
- [11] B. D. Williamson, J. Feng, Efficient nonparametric statistical inference on population feature importance using shapley values, in: *Proceedings of the 37th International Conference on Machine Learning (ICML)*, Vol. 119 of PMLR, 2020, pp. 10282–10291.
- [12] I. Covert, S. M. Lundberg, S.-I. Lee, Explaining by removing: A unified framework for model explanation, *Journal of Machine Learning Research* 22 (209) (2021) 1–90.
- [13] A. Catav, B. Fu, Y. Zoabi, A. Weiss-Meilik, N. Shomron, J. Ernst, S. Sankararaman, R. Shamir, Marginal contribution feature importance – an axiomatic approach for explaining data, in: *Proceedings of the 38th International Conference on Machine Learning (ICML)*, Vol. 139 of PMLR, 2021, pp. 1324–1335.
- [14] M. Grabisch, M. Roubens, An axiomatic approach to the concept of interaction among players in cooperative games, *International Journal of Game Theory* 28 (4) (1999) 547–565.
- [15] M. Sundararajan, K. Dhamdhere, A. Agarwal, The shapley taylor interaction index, in: *Proceedings of the 37th International Conference on Machine Learning (ICML)*, Vol. 119 of PMLR, 2020, pp. 9259–9268.
- [16] L. Breiman, Random forests, *Machine Learning* 45 (1) (2001) 5–32.
- [17] A. Fisher, C. Rudin, F. Dominici, All models are wrong, but many are useful: Learning a variable’s importance by studying an entire class of prediction models simultaneously, *Journal of Machine Learning Research* 20 (177) (2019) 1–81.

- [18] O. Ivanov, M. Figurnov, D. Vetrov, Variational autoencoder with arbitrary conditioning, in: Proceedings of the 7th International Conference on Learning Representations (ICLR), New Orleans, USA, 2019.
- [19] Y. Li, S. Akbar, J. B. Oliva, ACFlow: Flow models for arbitrary conditional likelihoods, in: Proceedings of the 37th International Conference on Machine Learning (ICML), Virtual Event, 2020, pp. 5831–5841.
- [20] J. You, X. Ma, D. Y. Ding, M. Kochenderfer, J. Leskovec, Handling missing data with graph representation learning, in: Advances in Neural Information Processing Systems 33 (NeurIPS), Virtual Event, 2020, pp. 19075–19087.
- [21] C. Cremer, X. Li, D. Duvenaud, Inference suboptimality in variational autoencoders, in: Proceedings of the 35th International Conference on Machine Learning (ICML), Stockholm, Sweden, 2018, pp. 1078–1086.
- [22] B. Amos, Tutorial on amortized optimization, Foundations and Trends in Machine Learning 16 (5) (2023) 592–732.
- [23] N. Jethani, M. Sudarshan, I. Covert, S.-I. Lee, R. Ranganath, Fast-SHAP: Real-time Shapley value estimation, in: Proceedings of the 10th International Conference on Learning Representations (ICLR), Virtual Event, 2022.
- [24] I. Covert, C. Kim, S.-I. Lee, Learning to estimate Shapley values with vision transformers, in: Proceedings of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023.
- [25] L. Bertinetto, J. F. Henriques, P. H. S. Torr, A. Vedaldi, Meta-learning with differentiable closed-form solvers, in: Proceedings of the 7th International Conference on Learning Representations (ICLR), New Orleans, USA, 2019.
- [26] K. Lee, S. Maji, A. Ravichandran, S. Soatto, Meta-learning with differentiable convex optimization, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, USA, 2019, pp. 10657–10665.
- [27] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, in:

Advances in Neural Information Processing Systems (NeurIPS), Vol. 30, 2017.

- [28] V. Vovk, A. Gammerman, G. Shafer, Algorithmic learning in a random world (2005).
- [29] J. C. Harsanyi, A simplified bargaining model for the n-person cooperative game, International Economic Review 4 (2) (1963) 194–220.